Interpretable Rule-Based Fuzzy ELM and Domain Adaptation for Remote Sensing Image Classification

Saroj K. Meher, *Senior Member, IEEE*, and Neeta S. Kothari, *Graduate Student Member, IEEE*

***Abstract*— The size of the remote sensing (RS) data is increasing exponentially with modern technological evolution. Efficient mapping with these data using a supervised model demands ample labeled samples, and obtaining them is expensive and time-consuming. For that reason, domain adaptation (DA) methodologies that work with limited available knowledge in terms of labeled samples or trained models are highly successful. DA uses the model trained for a domain called "source" and aims to classify the data of a related domain called "target." However, an insufficient amount of labeled samples in the target domain data set is still a bottleneck for DA models used for RS images. To address this, we have proposed an efficient DA classification model using an interpretable rule-based fuzzy extreme learning machine (IRF-ELM). These rules are derived using the maximum fuzzy membership value of features characterized by class-belonging fuzzification and two rule extraction matrices. A partially connected ELM architecture is then designed with these rules. The fuzzification process typically aligns the target domain's feature space to the source domain in the best possible ways to transfer knowledge between them. The proposed model's superiority to similar other methods is verified using different RS data sets in terms of various performance measurement indexes.**

***Index Terms*— Extreme learning machine, domain adaptation, fuzzy inference system, land cover classification, remote sensing.**

I. Introduction

ADVANCEMENT in the Earth observation systems generates a significant amount of quality remote sensing (RS) image data and makes them available with different spectral, spatial, and temporal resolutions. These data open up several opportunities for various tasks, e.g., repetitive monitoring of Earth's surface, dynamic process monitoring, large-scale image processing, land resource management, and land use planning [1]. RS image classification is one of the bases for many such environmental and socioeconomic applications. There are two main types of classification techniques, and the advantage of the supervised approach is the use of labeled data in the modeling process. However, supervised classification of RS images faces various challenges, such as limited labeled samples, spectral heterogeneity of the objects, and intraclass variability in the detailed information. Among them, the scarcity of labeled examples poses the challenge in estimating the class conditional probability density function of a classification model.

Moreover, the collection of large-sized labeled samples in the RS domain is highly expensive and time-consuming, not to mention that the targets' variability tends to make them useless. Semisupervised classification models are successfully used for such scenarios but pose the challenges of assuring that a significant relationship is labeled and unlabeled samples' data distribution. Several attempts have been made to mitigate these and exploring in the direction of domain adaptation (DA) [2] for land cover classification of RS images.

DA methodology aims to use the acquired existing knowledge of a domain (i.e., source) for learning the model and performs the classification operation on a new, but related domain (i.e., target). This approach's motivation is inspired by the human problem-solving ability to utilize previously acquired knowledge to solve a new but related/similar domain problem. In general, DA methods are broadly categorized into three approaches [2]: supervised, semisupervised, and unsupervised. These categorizations are based on the availability and use of labeled samples of source and target domains. Large-sized labeled examples of target domain are not used both in unsupervised and semisupervised DA approaches, and hence, the outcome becomes more error-prone. However, the supervised DA method effectively uses the available label information without satisfying many assumptions and conditions.

With the advances in supervised DA models for RS image classification, various training strategies are adopted, as done in [2], [3]. These are broadly categorized into four groups, i.e., based on invariant feature selection, classifier adaptation, adaptation through active learning, and data distribution adaptation [2]. Most of these research works have been focused on data distribution adaptation or heterogeneous DA strategies. Here, both the domains ensure their mapping to a shared space through the process of making data distribution of domains as similar as possible. Because RS data are mostly available from different sensors in large sizes, we have considered the adaptation strategies of data distribution for designing a supervised DA-based classification model.

With the adaptation method, the information contained in the source and target domains needs to be explored to find the commonality in their distributions. In this direction, the fuzzy

Manuscript received September 7, 2020; revised September 14, 2020; accepted September 14, 2020. Date of publication October 7, 2020; date of current version June 24, 2021. *(Corresponding author: Saroj K. Meher.)*

The authors are with Systems Science and Informatics Unit, Indian Statistical Institute, Bangalore Centre 560059, India (e-mail: saroj.meher@isibang.ac.in; sharmaneeta31@gmail.com).

Digital Object Identifier 10.1109/TGRS.2020.3024796

1558-0644 © 2020 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:05:59 UTC from IEEE Xplore. Restrictions apply.

logic-based approach can be a better choice because of its ability to conceptualize the impreciseness present in data sets and visualize the overlapping of impreciseness. Keeping this aspect in mind, we have fuzzified the input feature spaces of both domains using the same fuzzy membership functions (MFs) and developed a certain extent of overlapping in data sets' impreciseness. This preprocessing of data is advantageous to carry out heterogeneous supervised DA to get better classification accuracy.

In another computing paradigm, feedforward NNs (FFNNs) have been extensively studied and applied for classification and regression tasks. Since their inception, they have undergone different ways of improvements in terms of architectures and learning algorithms. Over the past years, FFNNs are identified as the most popular models for classification, which provides good classification accuracy and approximate any function for a finite number of training samples, using the gradient descent learning algorithms, such as backpropagation. However, the local minima issue and strategic optimization of network parameters make the FFNN as a slow learner and by-chance-based decision-maker. To overcome these issues, Huang *et al.* [4] proposed a single-hidden-layer-based FFNN called extreme learning machine (ELM), where the input weights are randomly initialized, and output weights between the hidden and output layers are analytically determined with the Moore–Penrose (MP) generalized inverse method. ELM has, thus, emerged as a simple, effective, and efficient algorithm for the pattern classification task. To obtain the improved versions, original ELM has undergone many variations concerning modeling, associated algorithms, and applicability.

To the best of our knowledge, there is not much exploration of the ELM-based DA approach for land cover classification of RS image. Most of the ELM-based DA classification methods lack in the design of likely optimum network architecture and make them more complicated with the increase in hidden-layer neurons of the ELM. We have attempted to make the ELM network architecture simple, interpretable, and computationally efficient. We have formulated an efficient method of deriving fuzzy rules from the fuzzified input feature space and accordingly designed the ELM network in this direction. A fuzzy inference system (FIS) is mostly clubbed with the ELM network architecture to formulate an interpretable and efficient network architecture. For the supervised classification task, Wong *et al.* [5] successfully incorporated the fuzzy concept in the form of fuzzy membership values and fuzzy rules for the ELM. The network is computationally efficient, and the authors have claimed the interpretability of the ELM network. However, the network architecture depends on randomly generated information in the form of two matrices that do not justify the true interpretability. Also, the authors have taken fixed parameters for fuzzy MFs regardless of the data sets, and thus, the method does not appropriately reflect the data distribution. To improve this ELM architecture's performance and make it truly interpretable based on the study's data, we have used the data-dependent fuzzification approach and the corresponding strategic rule generation method. More importantly, the proposed model preserves the ELM architecture's randomness aspect between the input and hidden layers.

With all these motivations, we have introduced an interpretable rule-based fuzzy ELM (IRF-ELM) classification model for RS images in a DA framework. The model uses four steps of contributed operations. The first one is applying data-dependent fuzzification for minimizing the data distribution dissimilarity between the source and target domains. Second, the maximum fuzzy membership value-based rule extraction using the source–domain data develops an IRF-ELM architecture. The third contribution is using element values of the rule matrices (created with these extracted rules), directly as the ELM network's weight values between the input and hidden layers. Finally, we have applied the IRF-ELM framework in the land cover classification of RS images using three DA algorithms, such as DA with parameter transfer (DAPT), source DAELM (DAELM-S), and target DAELM (DAELM-T). We trained the proposed model with very few labeled samples of the target domain and more samples of the source domain to classify the entire target domain samples. The proposed model's efficacy is compared with various ELM-based DA methods in terms of different measurement indexes.

II. Literature Review

Remotely sensed images have become a valuable source of information for many real-life applications [1]. However, the task of mapping using a supervised classifier faces severe challenges. In this direction, DA methods have played a crucial role [2]. In the recent past, various supervised DA methodologies [6], [7] have been proposed that perform the task more effectively than the semisupervised and unsupervised approaches.

There exist four different categories of DA approaches [2]. Among them [7] and [8], the data distribution adaptation approach is the most popular and successful one. As suggested by Tuia *et al.* [2], the heterogeneous DA strategy [9] is suitable for RS data sets acquired by different sensors. Using the data distribution adaptation strategy, recently, Zuo *et al*. [10] described a DA method based on fuzzy transfer learning (TL) using an infinite Gaussian mixture model and active learning. The authors have considered a deterministic grouping of training samples using the fuzzy C-means algorithm and estimated the centers of fuzzy MFs for rule extraction. A fuzzy rule defines the relationship between the fuzzified input feature and output classes. Several research efforts have been made to extract meaningful fuzzy rules [11]–[14] for supervised classification tasks. Various studies [5], [11]–[14] have been made to incorporate fuzzy rules with ELM architecture for supervised classification.

Intuitively, it makes sense to note that the fuzzification process [15] can provide a platform for data structure overlapping between the two DA method domains. In this study, we have considered two methods of fuzzification, i.e., class-nonbelonging (CNB) and class-belonging (CB), which are similar to the methods proposed in [16] and further adapted successfully in [14], [17], and [18].

In another aspect of the computing paradigm, NN [19] has proved its effectiveness, particularly the single-layered FFNN,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:05:59 UTC from IEEE Xplore. Restrictions apply.

but suffered from many bottlenecks, such as long training time, black-box phenomenon, and generalization ability. These issues are mostly addressed by the ELM [4], which works with the one-pass method of training samples. Several attempts have been made [4], [20] to address the classification problems using the ELM. In a survey paper on ELM-based DA algorithms, Salaken *et al.* [21] discussed the weighted ELM-based TL for the RS image classification. The article described the method of obtaining new output weights by minimizing a cost function over the target domain samples using an error function that is attributed to the connection weights. However, the method suffers from the overfitting issue because of the excess use of model parameters. Zhang and Zhang [6] proposed a unified framework called DA extreme learning machine (DAELM; named DAELM-S and DAELM-T), which extended the ELM to handle DA problems for improving the transferring capability of ELM. However, the method becomes ineffective when the distribution differences between the domains are large. Another application of employing ELM for DA (named DAPT) in RS, based on the transfer of ELM parameters [7], could effectively address the data distribution difference between the domains and surpassed the performance of DAELM-S and DAELM-T approaches. Keeping all these aspects in terms of merits and demerits, we have introduced an IRF-ELM classification model using these three DA frameworks for RS images.

Performance comparison and analysis of the IRF-ELM model using the DAPT-based DA algorithm is made with five fuzzy and four nonfuzzy DA methods. The fuzzy models are fuzzy refinement DA (FR-DA) [22], fuzzy-ELM (F-ELM), i.e., on the equivalence of FISs with DAPT DA algorithm (F-ELM-DAPT) [5], the DA model based on the Takagi–Sugeno–Kang fuzzy system (TSK-DA) [23], and the fuzzy DA with infinite Gaussian mixture model and active learning (IGMM-AL-DA) [10]. The four nonfuzzy models are conventional ELM with DAPT-based DA (ELM-DAPT) algorithm [7], autoencoder-based DA (AE-DA) [24], geodesic flow kernel DA (GFK-DA) [25], and DA via transfer component analysis (TCA-DA) [26]. In addition, we have compared the proposed model with adversarial discriminative DA (ADDA) model [27] using LeNet-5 convolutional NN (CNN) architecture [28].

We have used eight indexes for performance evaluation, such as percentage of overall accuracy (OA), precision, recall, kappa coefficient (KC) [29], dispersion score (DS) [18], β index [30], the Davies–Bouldin index (DB index) [31], and computational time T_c.

III. Proposed Methodology

This article proposes an IRF-ELM classification model for RS images in DA's framework. The training phase of the model has four steps of operation, as shown in Fig. 1. The network architecture is designed with the first three operations. In the first operation, the original features of the RS image in the source domain are fuzzified with a CB fuzzification method for better interpreting the information in the data set. The second operation extracts the most relevant and informative fuzzy if-then rules from these fuzzified features. These rules extract

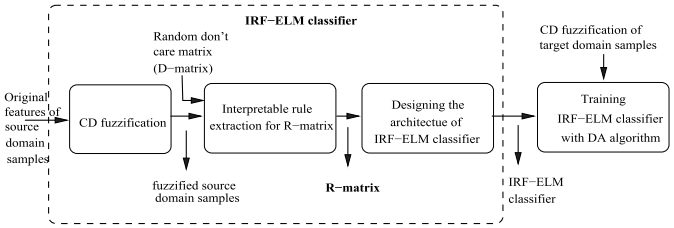


Fig. 1. Schematic flow diagram of the training stage of DA model with the proposed IRF-ELM network for RS image classification.

the data's knowledge and establish a relationship between the fuzzified input features and output class labels. For a fuzzy rule, we consider each feature's most significant membership value as the premise and corresponding class label as the consequent part. We have represented these rules with a binary 2-D rule matrix (R-matrix), further refined by a randomly generated don't care matrix (D-matrix). The third operation in Fig. 1 designs the architecture of the IRF-ELM model with these derived fuzzy rules. Once the IRF-ELM model's architecture is designed, it is then trained with the DA algorithm using the fuzzified features of the samples of source and target domains, as shown in the fourth step of Fig. 1. We have used the same fuzzy MF for fuzzyfing both source and target domain samples to obtain a common platform for their data distribution. To train the IRF-ELM model, we have considered three different ELM-based DA algorithms, such as DAELM-S, DAELM-T, and DAPT, and described their operational steps in the following.

Please note that both source and target domain data are multispectral RS images and contain four spectral information bands. Target data [linear imaging self-scanning system (LISS-II)] have 0.45–0.86-μm spectral variation of wavelength with a 36.25-m spatial resolution, whereas source data (LISS-III) have a spectral variation of wavelength 0.52–1.70 μm with a 23.5-m spatial resolution. In summary, both the domain data are of different feature spaces and spatial resolutions. A detailed description of these data sets is given in Section V.

A. Fuzzification of Input Features

Fuzzy logic is an appropriate tool to handle vagueness and uncertainty present in data sets, as in RS imagery. Fuzzification with different fuzzy sets plays a vital role in exploring the feature space of RS image for improved interpretation of information. To transform the original input feature spaces of both source and target domains into common fuzzified feature spaces in the DA platform, we have used the same fuzzy parameters of the MF of a fuzzy set. This fuzzification strategy helps address the distribution-dissimilarity between the domains effectively and amicable for DA methodologies-based classification models. Section III-A1 describes the CB and CNB methods of fuzzification briefly.

1) Class-Nonbelonging Fuzzification: Each input feature is characterized by a linguistic variable corresponding to a fuzzy set. The number of linguistic variables needs to be chosen for the best description of the data set. Hence, we have used three

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:05:59 UTC from IEEE Xplore. Restrictions apply.

linguistic variables, i.e., low, medium, and high corresponding to three fuzzy sets. Thus, the n-dimensional feature space ($\mathbf{F} = \mathrm{F}_1, \mathrm{F}_2, \ldots, \mathrm{F}_n$) is transformed into $(3 \times n)$-dimensional space with CNB fuzzification.

2) Class-Belonging Fuzzification: Class discriminative information of each feature is extracted using the CB fuzzification method, which depends on the number of classes present in the data set. With CB fuzzification, each feature of a sample of a data set with C number of classes is characterized into C number of fuzzified features. Thus, an n-dimensional feature space is represented by $(n \times C)$-dimensional fuzzified feature space using the CB approach. We have used π-type MF for this fuzzification and is given as

$$\begin{aligned}\mu(F_n; a, r, b) &= 0, \quad F_n \le a \\ &= 2^{N-1}[(F_n - a)/(r - a)]^N, \quad a < F_n \le p \\ &= 1 - 2^{N-1}[(r - F_n)/(r - a)]^N, \quad p < F_n < r \\ &= 1 - 2^{N-1}[(F_n - r)/(b - r)]^N, \quad r < F_n < q \\ &= 2^{N-1}[(b - F_n)/(b - r)]^N, \quad q < F_n < b \\ &= 0, \quad F_n \ge b \end{aligned} \tag{1}$$

where N is the fuzzifier, r is the center point, and p and q are the two crossover points of the MF. The values of p, q and r are estimated as $p, q = \mathrm{mean}(F_n) \pm [\max(F_n) - \min(F_n)]/2$, and $r = \mathrm{mean}(F_n)$, where mean, min, and max are mean, minimum, and maximum values of feature F_n of the data set. Using the training data, membership values of samples are determined. The value becomes closer to 1 or 0.5 depending on its position. Exactly at the center and crossover points, the maximum membership value of 1 and a minimum value of 0.5 are obtained. Accordingly, the classwise feature's membership is found out for each sample, and hence, the process becomes a data-dependent and CB fuzzification. The CB fuzzified features are, thus, represented as

$$\begin{aligned}F_{\mathrm{CB}} = [&\mu_1(F_1), \mu_2(F_1) \ldots \mu_c(F_1) \ldots \mu_C(F_1), \\ &\mu_1(F_2), \mu_2(F_2) \ldots \mu_c(F_2) \ldots \mu_C(F_2), \ldots \\ &\mu_1(F_n), \mu_2(F_n) \ldots \mu_c(F_n) \ldots \mu_C(F_n)]. \end{aligned} \tag{2}$$

In the CB fuzzification using π-type MF, the minimum and maximum values of feature F_n are considered two boundary points of the data set, hence denoted as the crossover points.

In this study, DA methodologies are applied to the two RS data sets collected from two related domains. These two data sets have the same type and number of classes with variations of wavelengths and sensor channels. Because the CB fuzzification process can be more advantageous to bring the data distribution of both source and target domains closer, the proposed classification model has used this method to fuzzify the input features.

B. Fuzzy Rule Extraction to Generate Rule Matrices

Fuzzy rules represent the domain knowledge obtained from the training data. A rule derives the relationship between the fuzzified input feature and output classes. Recently, Wong *et al.* proposed a method (F-ELM) for fuzzy rule generation and designed an ELM-based classification model.

TABLE I

MAXIMUM FUZZY MEMBERSHIP VALUE-BASED RULE EXTRACTION

Sample	MF								
	(F_1)			(F_2)			(F_3)		
	$\mu_{c1}(.)$	$\mu_{c2}(.)$	$\mu_{c3}(.)$	$\mu_{c1}(.)$	$\mu_{c2}(.)$	$\mu_{c3}(.)$	$\mu_{c1}(.)$	$\mu_{c2}(.)$	$\mu_{c3}(.)$
1	0.013	0.055	**0.986**	0.760	0.426	**0.893**	0.771	0.632	0.836
2	0.347	0.944	0.652	0.500	0.893	0.704	**0.998**	0.886	0.709
3	0.125	0.500	**0.875**	0.855	0.189	**0.952**	**0.971**	0.632	0.836
4	0.222	0.777	**0.778**	0.357	0.988	0.573	**0.998**	0.959	0.632
5	0.055	0.222	**0.944**	0.760	0.426	**0.893**	**0.971**	0.632	0.836

The technique takes a fixed number of fuzzy parameters for fuzzifing input features, equivalent to a FIS, and provides good classification performance in the supervised framework. The authors have used a random rule generation strategy. However, the random approach compromises the interpretability aspect of rules and fails to reflect the training data's actual knowledge. We have reduced the random aspects in the rule extraction operation and proposed a strategic way using the data-dependent approach for designing a supervised DA-based classification model. In this method, the fuzzy set's maximum fuzzy membership value is used for deriving the rule. A brief description of this method is made in the following with an example data set.

Initially, the number of rules (= number of nodes in the hidden layer of ELM) to be extracted is decided for a given data set. For example, to generate H rules, the whole samples of the training set are partitioned into H subsets of equal size. One subset is used to extract one rule, and thus, the rule acts as a representative of that set of samples. Let us consider an example data set of 30 samples with three features and three classes. Let $F_i = [F_{i1}, F_{i2}, F_{i3}]$ be the ith pattern/sample. These three features of F_i are then transformed into nine (3×3) CNB fuzzified features. For this data set, let $H = 6$ (number of rules), and accordingly, the whole data set is divided into six subsets with five samples in each subset. Samples of one such subset with their CNB fuzzified features are shown in Table I. It is observed from Table I that the majority of samples are having the maximum membership values for fuzzy set *c3* of feature F_{i1}, *c3* of F_{i2} and *c1* of F_{i3}. Hence, these elements together derive one rule that represents this subset of five samples. In this manner, the rest five rules are extracted from the remaining five subsets of samples.

The derived rules are represented in the form of a binary 2-D rule matrix (R-matrix), as shown in Fig. 2. The row of the R-matrix in Fig. 2 is equal to the number of fuzzified features (here nine), and the columns are equal to the number of rules (here six). The element value of a column (rule) is assigned "1" if the corresponding fuzzified features have maximum membership values for most of the samples. The remaining elements of that rule are assigned the value "0." For example, the values of elements in the first rule (i.e., first column) of R-matrix (see Table I) are assigned as $R(3, 1) = 1$, $R(6,1) = 1$, and $R(7, 1) = 1$ and others as 0. Furthermore, the R-matrix is refined by a don't care matrix (D-matrix) to remove the rule's insignificant features. D-matrix is a 2-D randomly generated binary matrix of size $F \times R$, where F is the number of original features, and R is the number of fuzzy rules. For example, in a D-matrix with $D(3, 1) = 1$,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:05:59 UTC from IEEE Xplore. Restrictions apply.

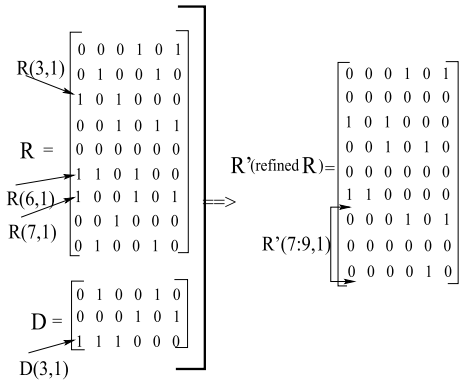


Fig. 2. Illustration of R, D, and R'(refined) matrixes for an example data set.

it implies that feature 3 (rows 7–9 of R-matrix) is redundant in rule 1 (column 1 of R-matrix). Thus, in rule 1, all elements for feature 3 in R-matrix are reset to 0, i.e., $R'(7{:}9, 1) = 0$. This rule extraction strategy serves three purposes. These are efficient representations of domain knowledge, the fulfillment of the requirement of a large number of rules (hidden nodes) by dividing the whole set of samples into subsets, and maintaining the randomness aspect of ELM weight parameters using D-matrix.

C. Designing the Architecture of IRF-ELM

In this study, we have enhanced the conventional ELM and F-ELM knowledge extraction ability and developed the IRF-ELM classification model. This section discusses the basics of ELM and designs the IRF-ELM model.

1) Extreme Learning Machine: A brief description of the functionalities of ELM is given in the following. The learning coefficients between the input and hidden layers are taken randomly in the ELM algorithm, and the coefficients between the hidden and output layers are determined analytically using the MP generalized inverse.

Let K be the training set of N samples with d features and l labels and is represented as $K = \{(x_i, y_i) |\ x_i \in \mathbb{R}^d, y_i \in \mathbb{R}^l,\ i = 1, 2, \ldots, N\}$, where $x_i = [x_{i1}, x_{i2}, \ldots, x_{id}]^T$ and $y_i = [y_{i1}, y_{i2}, \ldots, y_{il}]^T$. The output function of SFFNN with L hidden nodes and activation function $f(x_i)$ for an input x_i is given as

$$f_L(x_i) = \sum_{j=1}^{L} \beta_j \cdot f(w_j \cdot x_i + b_j) = y_i, \quad i = 1, 2, \ldots, N. \tag{3}$$

The final hidden node matrix is the successive arrangement of all the outputs of hidden nodes. w_j in (3) is an d-dimensional weight vector that connects jth hidden node to input nodes. The objective of ELM is to approximate N samples with L hidden nodes, and thus, the solution for w_j, b_j, and β_j is expected to exist for completely zero prediction error at the network output. Therefore, the compact form of (3) becomes $\mathbf{H}\widehat{\beta} = Y$, where $\mathbf{H}(w_1, w_2, \ldots, w_L, b_1, b_2, \ldots, b_L, x_1, x_2, \ldots, x_N)$ is the matrix form of the hidden layer output of the network and $h_{ij} = f(w_j.x_i + b_j)$ is the output of jth hidden node for x_i, $j = 1, 2, \ldots, L$ and $i = 1, 2, \ldots, N$.

$\widehat{\beta} = [\beta_1, \beta_2, \ldots, \beta_L]^T$ and $Y = [y_1, y_2, \ldots, y_L]^T$ are the output weight matrix and the actual/target matrix, respectively. In real-time applications, it is expected that the training error is never equal to zero for $N \gg L$. For getting the error ≈ 0, Huang *et al.* [4] proposed to assign the values of w_j and b_j randomly to make the system possibly linear, in order to estimate the output weights as $\beta = \mathbf{H}^{\dagger}Y$, where $\mathbf{H}^{\dagger}$ is the MP generalized inverse matrix of $\mathbf{H}$. Thus, the ELM algorithm for K training set [see (3)] can be executed in three steps: 1) randomly assign the parameters w_j, b_j with $j = 1, 2, \ldots, N$; 2) calculate $\mathbf{H}$; and 3) find the output weight matrix using $\beta = \mathbf{H}^{\dagger}Y$.

2) IRF-ELM Classification Model: The IRF-ELM model's architecture is a partially connected ELM, where the connections between the input and hidden layers are made based on fuzzy if-then rules. Each of the derived fuzzy rules is meant for one node of the hidden layer. Therefore, the number of rules is equal to the number of hidden nodes selected for the IRF-ELM model. We have used the binary R-matrix (as described in Section III-B) directly as the input weight matrix between the input and hidden layers of IRF-ELM, where the values of weights are either "0" or "1." The "0" valued weight implies no connection between the nodes, and the "1" valued weight means a connecting link with unity weight. Unlike the conventional ELM, where the input to hidden layer weights is randomly initialized, we have used knowledge-based informative and binary weights. However, the randomness aspect of an ELM is maintained using the D-matrix. Similar architecture but with more computational complexities is described in F-ELM, where both R-matrix and D-matrix are random. The fundamental difference between the proposed and the method described in F-ELM is that we have maintained the randomness aspect of ELM and, at the same time, incorporated the interpretable domain knowledge in it, unlike the method in F-ELM.

For illustration, the IRF-ELM model's architecture for a data set with three input features, three output classes, and six fuzzy rules (derived from the training samples) is shown in Fig. 3. The model has three, six, and three nodes in the input, hidden, and output layers. Fig. 3 shows an example of the network architecture of an IRF-ELM using the R-matrix and D-matrix (see Fig. 2).

D. Training the IRF-ELM Model With DA Algorithms

DA algorithms aim to solve particular domain data classification tasks with very few labeled samples. Various studies have explored the practical and straightforward characteristics of ELM and use it in the DA framework. Zhang and Zhang [6] proposed a unified framework called DAELM based on a conventional ELM for drift compensation in the E-nose system. The authors proposed two algorithms called DAELM-S and DAELM-T in the unified framework. These algorithms train an ELM classifier with a limited and sufficient number of labeled samples obtained from a target and source domains, respectively. The algorithms explored the new

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:05:59 UTC from IEEE Xplore. Restrictions apply.

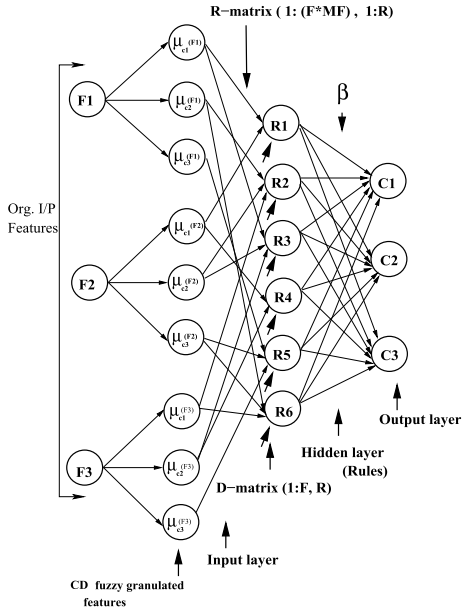


Fig. 3. IRF-ELM architecture.

perspective of ELM for DA and outperformed traditional kernel-based and manifold-regularization-based DA methods. Another ELM-based DA algorithm called DAPT was proposed for RS image classification. In this study, the ELM model parameters obtained from the target domain are transferred into the source–domain's parameter space to bring the classifiers' parameter distribution closer. The performance of DAPT is improved over several DA methods, including SVM-based, DAELM-S, and DAELM-T. This study has considered DAELM-S, DAELM-T, and DAPT-based DA algorithms for training the proposed IRF-ELM model for land cover classification of RS image. The training and testing processes of the DA-based IRF-ELM for RS image classification are illustrated in Algorithm 1.

Algorithm 1 Algorithmic Steps of the DA Model With the Proposed IRF-ELM Classification Model

1) Given: Source domain labeled samples $\{X_{s_l}, l_{s_l}\} = \{x^i_{s_l}, l^i_{s_l}\}_{i=1}^{S_{s_l}}$ where $l^i_{s_l}$ is the class label for the sample $x^i_{s_l}$. Target domain labeled labeled samples $\{X_{t_l}, l_{t_l}\} = \{x^j_{t_l}, l^j_{t_l}\}_{j=1}^{S_{t_l}}$ where $l^j_{t_l}$ is the class label for the sample $x^j_{t_l}$. Target domain unlabeled samples $\{X_{t_u}\} = \{x^k_{t_u}\}_{k=1}^{S_{t_u}}$.
2) Fuzzification of the input features of source domain labeled samples using CB method.
3) Extract rules from the source domain labeled samples (S_{s_l}) to generate the R-matrix.
4) Randomly generate don't care matrix (D-matrix) and update/refine the R-matrix to R'-matrix.
5) Initialize the weight parameters between the input to hidden nodes of IRF-ELM network with binary weight values stored in the R-matrix and hidden bias B.
6) Calculate the output matrix $\mathbf{H}_{s_l}$ (for labeled source samples), $\mathbf{H}_{t_l}$ (for labeled target samples) and $\mathbf{H}_{t_u}$ (unlabeled target samples) (Eqn. (3)).
7) Compute β according to the DA algorithm (DAELM-S or DAELM-T or DAPT).
8) Calculate the predicted output $y^k_{t_u}$ as
   - For DAELM-S and DAELM-T: $y^k_{t_u} = \mathbf{H}^k_{t_u}.\beta$
   - For DAPT: $y^k_{t_u} = \mathbf{H}^k_{t_u}.W.\beta$

IV. Performance Measurement Indexes

Indexes are used to evaluate the performance of the proposed IRF-ELM classification model for RS images. Some of these parameters are computed from the confusion matrix (CM). CM is a cross-tabulation of agreement between the actual and estimated class labels of samples obtained through a classifier. A brief description of these measures is given in the following.

A. OA, Precision, Recall, KC, DS, Boxplot, and T_c

OA is the ratio of correctly estimated classes of samples and the total presented samples. The summation of the diagonal elements of CM gives the value of OA. The interpretation of CM from a particular class point of view is given by Precision and Recall. Precision determines the total number of correctly classified samples from the total classified samples for a class, whereas Recall provides the correctly classified examples that the classifier attributed to a particular category. Precision and Recall can be thought of as the measures of a classifier's exactness and completeness, respectively. Hence, a low value of Precision and Recall is not desirable. Overall evaluation of the classifier concerning the accuracy of the individual class is performed using KC. Another critical index for the classwise performance evaluation of the classifier is DS. It estimates the distribution of the classified samples among the classes. Thus, it enables us to analyze the overlapping characteristics of different class boundaries in the data set. DS is defined as

$$\mathrm{DS}_i = \left[1 - \left(\lambda_1 \frac{V_i}{\mathrm{V}_i^{\max}} + \lambda_2 \frac{Z_i}{C-1}\right)\right]$$

where $i = 1, 2, \ldots, C$ (number of classes), V_i is variance of the elements in the ith row of CM, $\mathrm{V}_i^{\max}$ is the maximum variance corresponding to the ith row, which is obtained when all samples are correctly classified to the ith class, and Z_i is the number of element(s) in the ith row of CM with zero value. The weights λ_1 and λ_2 are assigned such that $\lambda_1 + \lambda_2 = 1$. It is to be noted that a higher value of OA, Precision, Recall, and KC and lower value of DS are desirable. One more index termed boxplot is used for the statistical significance of accuracies obtained with different models. The overall time required to train a model and test it is the computation time T_c (in seconds).

B. Beta Index and Davies–Bouldin (DB) Index

β index and DB index assess the segmentation quality. β index estimates the proportion of total variation over within-class variation. A good quality cluster must have high intercluster separation and intracluster compactness and homogeneity, which is measured using the DB index.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:05:59 UTC from IEEE Xplore. Restrictions apply.

V. Data Set Used

In the present experimental study, we have used both completely and partially labeled RS data sets to validate the DA framework-based IRF-ELM classification model's potentiality. A completely labeled data set denotes that all of its samples are labeled. This data set category is randomly partitioned into different training and test sets for model development and testing. Accordingly, the measurement indexes, such as OA, Precision, Recall, KC, and DS, are used for the performance analysis of models. On the other hand, a partially labeled data set denotes that only a small collection of samples from the whole group is labeled. Other instances are unlabeled because of unavailable ground truths. For these data sets, the classification model is trained using the available labeled samples and tested with the whole samples. In such a scenario, the clustering indexes, such as β and DB, measure the classifier's performance.

In this study, three multispectral RS data sets are collected from different sensors. We have used LISS-III and LISS-II data sets for the DA framework, as the source and target domains, respectively. The experimental labeled data consist of a fixed number of samples collected from each land cover class. To train the DA-based classification model, we have used very few target domain samples and many source–domain samples, whereas, for a conventional supervised framework-based model, different sets of training and test samples are prepared from the data sets. A brief description of these data sets is made in the following.

A. LISS-II Multispectral RS Data

This multispectral RS image was taken by Indian remote sensing (IRS) satellite with an LISS-II sensor. The image covers the regions of Kolkata city in India. It contains pixels of size (512 $\times$ 512) with 0.45–0.86-μm spectral variation of wavelength and 36.25-m spatial resolution. It consists of four spectral bands, i.e., the blue band (band 1), green band (band 2), red band (band 3), and near-infrared band (band 4). For this data set, six major land cover classes are considered, such as pond or fishery water (PW), turbid water (TW), concrete area (CA), habitation (HAB), vegetation (VEG), and open spaces (OS).

B. LISS-III Multispectral RS Data

This image data set contained pixels of size (1024 $\times$ 1024) for the region around Kolkata city in India. It was obtained from LISS-III [32] sensor. There are four spectral bands, i.e., two from the visible spectrum (green and red) and two from the infrared spectrum (near-infrared and shortwave infrared). The spatial resolution and spectral variation over the wavelength are 23.5 m and 0.52–1.7 μm. Similar to LISS-II data, the same six land cover classes are considered for LISS-III data.

C. SATIMAGE RS Data

SATIMAGE data set is collected from an RS image with pixel size (2340 $\times$ 2340) and was generated from Landsat Multispectral Scanner with a spatial resolution of 80 m. The data samples used in this study are subarea of the image having six major land cover classes, namely red soil, cotton crop, gray soil, damp gray soil, soil with vegetation stubble, and very damp soil. The data set consists of 6435 samples with 36 features, but the database designer recommends only four features for the classification task. We have considered this data set for the conventional supervised classification model, similar to the LISS-II data set.

VI. Experimental Results and Discussion

The efficacy of the proposed IRF-ELM classification model is evaluated with DA and supervised frameworks for RS images. Evaluation of model performance and the corresponding analysis of results are also made for the scenarios of completely and partially labeled data sets. For this study, 1200 samples (200 for each of the six classes) are selected and considered as the labeled data from each domain. We have randomly chosen 50% and 5% samples (equal percentage from each category) of source and target domains, respectively, and used as the model's training set. Once the classifier is trained, it is then tested on all 1200 samples of the target domain data set under the completely labeled scenario classification. We have also tested the model for the classification of the entire LISS-II image of sample size (512 $\times$ 512) under the partially labeled scenario classification.

With the different number of hidden nodes in ELM, a comparison of results using the model is made, but the critical analysis is presented for the likely optimum number of rules. For the comparative analysis of performance, we have designed models with different architectures, inputs, and rules. A brief description of these models is made in the following.

A. Model Description

We have developed five classification models to classify the RS images in both supervised and DA frameworks. These models differ concerning input fuzzification of features, rule extraction methods, and network architectures. Those are as follows: Model 1: Conventional ELM with original input features; Model 2: F-ELM with data-independent fuzzification; Model 3: F-ELM with data-dependent and CNB fuzzification of input; Model 4: F-ELM with data-dependent and CB fuzzification of input information; Model 4: Random rule-based fuzzy-ELM (RRF-ELM) with CB fuzzification of input; and Model 5: IRF-ELM with CB fuzzification of input data.

Model 1 is the conventional ELM with original input features. In F-ELM, the authors proposed a supervised classification method, which is adopted to develop Model 2. This model uses data-independent fuzzification of input features by taking five fixed center points with random parameters for the Gaussian MFs. These fuzzified features are used to compute F-ELM's hidden layer output in three stages that make the network architecture more complex. First, the function called modified probabilities-OR (MPRO-OR) is applied on the fuzzified input features to get the corresponding values for each of the hidden node/rule using the random rule matrix and the don't care matrix. Then, with the MPRO-OR function's results, each rule's firing strength is computed through a fuzzy

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:05:59 UTC from IEEE Xplore. Restrictions apply.

T-norm operator, and multiplication operation is performed similar to an FIS. In the next step, the hidden layer output matrix is obtained by combining all training samples' firing strengths. These steps make the network architecture of Model 2 more computationally complex. Furthermore, computing the outputs of hidden nodes becomes more burdensome because of the random weight matrix.

The proposed model in this study is designed for the possible improvement in Model 2. The three significant operations, i.e., input feature fuzzification, network design, and rule generation procedures, are modified to get the best possible performance over Model 2. First, the independent data fuzzification is replaced with data-dependent methods, such as CNB and CB. Model 3 and Model 4 have used these logics for fuzzifying the original features, keeping all other operational steps of Model 2 intact. CNB and CB are data-dependent fuzzification methods (see Section III-A) that work with the dependence of features to classes. The CB fuzzification has been proven better then CNB while analyzing Model 3 and Model 4. Therefore, we have considered the CB method of fuzzification for further improvement in Model 5. We have designed a partially connected three-layered ELM network architecture with a sigmoid activation function for the proposed model and calculated the error by the backpropagation algorithm. The interconnections between the input and hidden layers are based on the rule matrices (R and D) (see Section III-B).

For the proposed model, we have extracted the interpretable rules (binary R-matrix) from the source–domain training samples' data distribution. The concept of maximum fuzzy membership value is used for rule extraction. Finally, the proposed IRF-ELM model (Model 5) is developed with a three-layered fuzzy ELM using the CB fuzzified input features. The architecture is designed according to the interpretable rule matrix (R-matrix) that also serves as an input weight matrix. In the framework of DA methods (DAELM-S or DAELM-T or DAPT), the output of all models has been computed accordingly. The parameters in these DA algorithms are referred to in [6] and [7]. However, for the supervised framework of classification, the output weights between the hidden and output layers of the IRF-ELM network are determined analytically using MP pseudoinverse.

Separately, the performances of all five models are analyzed for each DA algorithm. Along with the DA platforms, these models' performance is analyzed for the conventional supervised classification platform. Performance of the proposed model (Model 5) is compared with other four models (Model 1, Model 2, Model 3, and Model 4) in terms of different measurement indexes, such as OA, T_c, Precision, Recall, DS, KC, β index, and DB index. Because of the random initialization of parameters in conventional ELM of Model (D-matrix and R-matrix in Models 2–5), each experiment is repeated ten times keeping other conditions/parameters intact. The average of these ten results is taken as the final result presented in Section VI.

In addition to these indexes, we have used the standard deviation of results for OA (OA_{std}) and KC (KC_{std}) to provide a measure of statistical significance. The statistical significance is also measured using the Box and Whisker Plot (or boxplot) analysis of OA results. All these critical analyses of results are made for all data sets and DA methods but presented for the DAPT-based DA classification algorithm. This is because a similar trend of performance is observed for the rest data sets and DA algorithms.

TABLE II
COMPARISON OF MODELS WITH DAPT ALGORITHM USING 50% SAMPLES OF LISS-III DATA AND 5% SAMPLES OF LISS-II DATA, AS TRAINING SET

Model	OA (in %) with number of rules				T_c (in sec.) with number of rules		
	200 rules	400 rules	600 rules	OA_{std} for 400 rules	200 rules	400 rules	600 rules
1	82.78	83.65	82.55	1.43	1.8652	2.9833	4.4988
2	85.93	86.13	85.84	1.38	1.8651	2.9824	4.4913
3	86.96	87.74	86.02	1.15	1.4824	3.1056	4.4599
4	88.08	89.28	88.27	0.90	1.3575	3.1435	4.5069
5	92.80	93.91	92.90	0.44	0.1551	0.2537	0.4121

B. DA Framework for Land Cover Classification of RS Image

This study considers two RS images (LISS-III and LISS-II) to analyze and verify the classification model with both DA and supervised frameworks. The SATIMAGE data set is only used for supervised framework-based classification. We have presented the classification results with relevant analysis and discussion using these data sets in the following. Classification of LISS-II data (target domain) is made with completely (with 1200 samples) and partially (with the whole image of size (512 × 512) pixels) labeled scenarios.

1) Classification With DAPT-Based DA Algorithm:

a) Completely labeled data scenario: Comparative analysis of all models in the classification of the LISS-II (target domain) data set is made with three DA algorithms. Classification results using these models in terms of OA and T_c with three DA algorithms, such as DAPT, DAELM-S, and DAELM-T, are presented in Tables II, V, and VI, respectively. These results are obtained by taking 50% samples (600 samples with 100 samples per class) of LISS-III data (source domain) and 5% samples (60 samples with ten samples per class) of LISS-II data (target domain), as the training set and the whole LISS-II data of 1200 samples as the testing set. We have considered three different numbers of rules for all these experiments, i.e., 200, 400, and 600. For justifying the statistical significance of performances, the standard deviation of OA (OA_{std}) over ten rounds of experiments (as mention earlier) is depicted in Section VI. The performance is also evaluated in terms of KC, KC_{std} (computed similar to OA_{std}), Precision, Recall, and DS using all three algorithms. However, we have presented and discussed these critical analyses of performance only with the DAPT algorithm because similar trends were observed with the others. With the DAPT-based DA method, the ELM parameters of the target are mapped to the source so that the target domain's classifier parameters are maximally aligned with the source.

Table II depicts the performance results in terms of OA and T_c, for all five models with DAPT-based DA algorithm. It is observed that Model 1, with the conventional ELM that takes the original input features, performed the worst

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:05:59 UTC from IEEE Xplore. Restrictions apply.

among the five models. This is obvious because the network uses the original features without any preprocessing. Again, the weights between the input and hidden layers are random, and no rules are used for knowledge extraction and designing network architecture. At the same time, Model 2, using F-ELM with data-independent fuzzification, provided better OA values of 85.93, 86.13, and 85.84 for 200, 400, and 600 rules in the corresponding ELM architectures, respectively, compared with Model 1. This improvement in the performance of Model 2 is attributed to the fuzzification process and rule extraction operation in the design of ELM architecture.

Furthermore, Model 3 with data-dependent and CNB fuzzification of input features provided 86.96, 87.74, and 86.02 for 200, 400, and 600 rules, which are better than Model 1 and Model 2. This shows the significance of the data-dependent and CNB fuzzification method for extracting better knowledge from the input data and helps the ELM network to derive more meaningful rules for improved classification accuracy. This accuracy is enhanced with Model 4 using all three numbers of rules in ELM, which uses the CB fuzzification method instead of CNB. This result signifies that the CB fuzzification method could derive improved class belonging/discriminatory information, unlike CNB, which became the reason for enhanced accuracy.

In the next step, we developed the proposed model (Model 5) using the IRF-ELM and the maximum fuzzy membership value-based rule extraction and CB fuzzification methods. The interpretable rules are derived from the source training samples and form a binary rule matrix (R-matrix). This R-matrix is used directly as an input weight matrix of ELM. In this way, the network architecture is simplified, and operations become straightforward. The improvement with Model 5 compared with others is visible from Table II. For example, with 200, 400, and 600 rules, Model 5 yielded the OA of 92.80, 93.91, and 92.90 compared with the values 88.08, 89.28, and 88.27, respectively, using Model 4. The improvement in the performance of Model 5[1] is about 4% better than Model 4, around 7% than Model 2, and nearly 11% better than the base model (Model 1), which is significant.

One important aspect that needs attention, i.e., all five models' accuracy, is the best with 400 rules compared with 200 and 600. It is also true that an increase in the number of rules/nodes in ELM from 400 to 600 does contribute to significant performance improvement, with the increase in computational burden. This is the same for all the models. This phenomenon can be seen from Table II using the computational time T_c. For Model 5, the T_c value with 400 rules is 0.2537 compared with 0.4121 for 600 rules. This improvement is almost two times bigger and that too, with similar accuracy for both rules. The reduction of T_c value in Model 5 is attributed to the network architecture and computation method. As discussed earlier, the architecture of F-ELM is complex. It has more stages of computation and, thus, exhibits more overall computation time.

[1]MATLAB code for Model 5 and data sets for source and target domains can be referred to DOI: 10.21227/v9v6-1197.

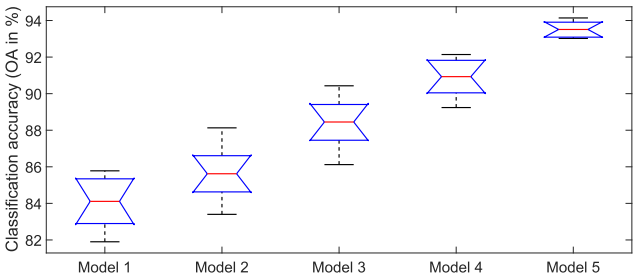


Fig. 4. Box plot of classification accuracies (OA) for different models.

On the other hand, the architecture of Model 5 is much simpler than F-ELM. Moreover, the input weight matrix is binary in Model 5, unlike conventional ELM (Model 1) and F-ELM, where it is random. The binary weight matrix lowers the computation time as it implies a connection with no calculation between nodes or no contact. Table II also shows the statistical significance of the models in terms of OA_{std}, in the classification of RS data using DAPT-based DA algorithm. It is seen that the importance of Model 5 is the highest (with low OA_{std} of 0.44) among the rest, which is reasonable and acceptable.

To analyze the ten rounds of experimental results using five models in terms of OA, we have constructed the corresponding boxplot and shown in Fig. 4. It is seen from Fig. 4 that the proposed Model 5 has shown encouraging results, among others, and the improvement is statistically significant. According to the boxplot interpretation of Fig. 4, the median line (red line in the box) for Model 1 is about 84%, and this value has increased to around 93% for Model 5. For the result points' dispersion, the interquartile ranges also obtained the box plot's superior value for Model 5. On average, all models' performance is obtained fairly, and no potential outliers are observed in the graph. In regards to the skewness, it is not straightforward and symmetrical, but it is better with Model 5 compared with others. Since the notches in the box plot do not overlap, it can infer that the true medians do differ and provide better confidence.

With all these observations, it can be inferred that the proposed model is a successful attempt to provide an efficient and straightforward solution for RS image classification in DA's framework. Also, this approach explored ELM to make it partially interpretable and partially connected network with a reduced computational burden.

The performance of models is then compared and critically analyzed (for 400 number of rules) with other measurement indexes, such as KC, KC_{std}, Precision, Recall, and DS, as shown in Table III. It is observed from Table III that the KC values increase from Model 2 to Model 5 and correctly align with the trend of results obtained with OA values. Along with this index, the statistical significance of results with KC_{std} also supported a similar improvement in the performance of models, as in the case of OA_{std}. This observation quantitatively rationalizes a better overall classwise agreement provided by the proposed IRF-ELM architecture-based model (Model 5). The DS values also supported the claim of superiority of models similar to Precision and Recall. These DS values are

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:05:59 UTC from IEEE Xplore. Restrictions apply.

TABLE III
COMPARATIVE ANALYSIS OF DA MODELS WITH OTHER INDEXES

Model	KC / (KC_{std})	Class	Training with 50% Source and 5% Target data		
			Precision	*Recall*	*DS*
2	0.808 / (0.583)	1	1.0000	0.5200	0.2890
		2	0.5505	0.6000	0.2890
		3	0.9898	0.9700	0.2390
		4	0.9707	0.9950	0.2390
		5	0.9604	0.9700	0.2890
		6	0.6800	0.9350	0.2890
5	0.903 / (0.0347)	1	0.9749	0.9700	0.1534
		2	0.9691	0.9400	0.2034
		3	0.9706	0.9900	0.1034
		4	0.9949	0.9800	0.1034
		5	0.9894	0.9350	0.2534
		6	0.9217	1.0000	0.0534

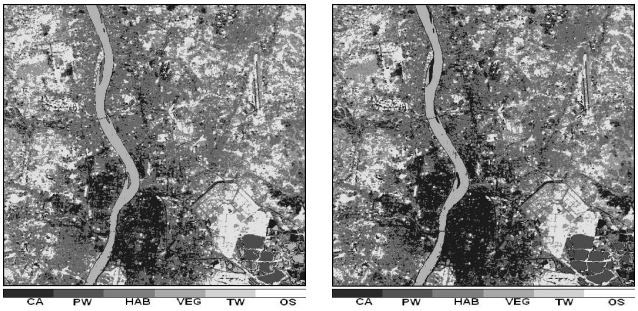
TABLE IV
PERFORMANCE OF MODELS WITH DAPT ALGORITHM FOR PARTIALLY LABELED SCENARIO USING LISS-II DATA

Training samples from Source/target domain	# of rules	β		DB	
		Model 1	Model 5	Model 1	Model 5
600 / 60	200	2.5716	6.3456	1.5671	0.9869
600 / 60	400	2.6753	7.6865	1.4782	0.8213
600 / 60	600	2.3717	7.4521	1.5678	0.9733

decreased from Model 2 to Model 5 for all classes. They indicate less dispersion of samples of a class, among other classes. These observations in terms of KC, Precision, Recall, and DS also hold with the other two DA algorithms (i.e., DAELM-S and DAELM-T), described in the next section.

b) Partially labeled data scenario: In this section, the proposed model's efficacy is evaluated for pixelwise land cover classification of the partially labeled RS data. With a partially labeled scenario, the classification is performed on the whole image, including the unlabeled samples/pixels. Similar to the classification of completely labeled data scenario, 600 samples (50% of total) of LISS-III source–domain data and 60 samples (5% of total) of LISS-II target domain data are taken as the training set. In terms of DB index and β index, a performance comparison between the conventional ELM (Model 1) and the proposed IRF-ELM (Model 5) model with the DAPT algorithm is only depicted in Table IV. This is because similar findings were obtained with the performance of Model 2, Model 3, and Model 4 and with DAELM-S and DAELM-T algorithms. To get the optimum number of rules in the ELM network, we tried and tested 200, 400, and 600 numbers and found 400 rules as the best one. It is observed with the classification results obtained by the proposed IRF-ELM model that β and DB values are the best among the results obtained with Model 1 and that too with any combination of the number of rules. In other words, β values of Model 5 is the higher, and the DB value is the lower, which clearly shows the superiority of Model 5.

Along with the quantitative evaluation and analysis of performance using β and DB indexes, we have also demonstrated the qualitative analysis of models. For qualitative evaluation, the classified image of the target domain (LISS-II) using F-ELM (Model 2) and the proposed IRF-ELM model



(a) (b)

Fig. 5. LISS-II image classified using (a) Model 1 and (b) Model 5.

TABLE V
MODEL COMPARISON WITH DAELM-S ALGORITHM USING 50% SAMPLES OF LISS-III DATA AND 5% SAMPLES OF LISS-II DATA, AS TRAINING SET

Model	OA (in %) with number of rules				Results for 200 rules-based network		
	200 rules	400 rules	600 rules	OA_{std} for 200 rules	KC	KC_{std}	T_c
1	82.93	82.08	82.31	1.908	0.7012	0.918	2.35
2	86.47	86.36	85.63	1.770	0.7784	0.603	1.62
3	88.11	88.02	87.89	1.463	0.8226	0.406	1.38
4	90.63	90.03	89.89	1.110	0.8567	0.413	1.39
5	93.26	93.11	92.76	0.874	0.9205	0.101	0.54

(Model 5) are shown in Fig. 5(a) and (b), respectively. The image obtained by Model 1 is not much clear, and different land cover regions are overlapped, which is not appealing visually. This haziness in the land cover regions is because of the result of an inferior classification of land covers by the model. On the other hand, the image obtained by the Model 5 is discrete, and different land cover regions are clearly segregated from each other, which can be visualized from Fig. 5(b). Thus, these analyses justify the superiority of the proposed classification model as more firm and reasonable in classifying RS images in DA's framework.

2) DAELM-S Algorithm-Based Classification of Completely Labeled Data Set: We have taken a few labeled target domain samples and more labeled source–domain samples to train the proposed IRF-ELM classification model with the DAELM-S algorithm. Like the DAPT-based model, we have used 60 (5% of total samples) and 600 (50% of whole samples) number of labeled examples for the target and source domains, respectively, for training the classifier. Table V illustrates the classification results of all the models with the DAELM-S algorithm. For the ELM architecture, we have used three different rules: 200, 400, and 600. However, better performance was observed (see Table V) using 200 rules compared with 400 and 600 numbers in all aspects. For that reason, the indexes, such as OA_{std}, KC, KC_{std}, and T_c, are depicted in Table V for only 200 rule-based architecture of ELM networks.

Similar to the DAPT-based DA algorithm, all models performed equivalently with the DAELM-S-based DA algorithm. For example, Model 1 performed the worst among all. This was expected because Model 1 used conventional ELM architecture. Its performance gradually increased using

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:05:59 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
MODEL COMPARISON WITH DAELM-T ALGORITHM USING 50% SAMPLES OF LISS-III DATA AND 5% SAMPLES OF LISS-II DATA, AS TRAINING SET

Model	OA (in %) with number of rules				Results for 200 rules-based network		
	200 rules	400 rules	600 rules	OA_{std} for 200 rules	KC	KC_{std}	T_c
1	83.63	82.60	82.11	1.810	0.7110	0.832	2.41
2	87.15	86.02	86.89	1.642	0.7846	0.617	1.66
3	88.24	87.23	87.84	1.305	0.8234	0.514	1.40
4	90.14	89.03	89.57	1.030	0.8617	0.304	1.37
5	93.25	92.80	92.01	0.724	0.9312	0.117	0.65

Models 2–5, and finally, the proposed IRF-ELM-based model (Model 5) performed the best. This observation noticeably justifies the significance of the data-dependent and CB fuzzification process that better exploits the input feature space than other fuzzy method-based models. Model 5 also takes advantage of the interpretable rule extraction method and partially connected network for enhanced accuracy. Besides, the T_c value is smaller for Model 5 compared with others. The KC values also support the superiority of Model 5 to others. The indexes OA_{std} and KC_{std} validate the statistical significance of these results. In all aspects, Model 5 yielded the encouraging performance with the DAELM-S-based DA method, which is similar to the results obtained with the DAPT-based DA algorithm.

3) DAELM-T Algorithm-Based Classification of Completely Labeled Data Set: The authors in the DAELM-T algorithm-based model have used unlabeled samples of the target domain in the training process of ELM. The algorithm is named DAELM-T. Following the DAELM-T algorithm, we have trained all the models and classified the target data. Table VI presents the classification results. With all models, the DAELM-T algorithm's application has shown a similar performance improvement trend as obtained with DAELM-S and DAPT algorithms. For example, the accuracy of the proposed IRF-ELM model (Model 5) is 93.25% with 200 rules, which is the highest accuracy obtained with all models and many rules. Computation time T_c is also very much reduced; model 5 took 0.653 s for classification with 200 rules, whereas Model 1 took 2.41 s. Model 2, Model 3, and Model 4 with the DAELM-T algorithm performed similarly with the DAELM-S algorithm except that there is a slight increase in accuracy in some cases. This slight increase in accuracy is attributed to the unlabeled samples of the target domain used during the models' training. Other findings and issues, as analyzed in the case of DAPT and DAELM-S-based DA algorithms, e.g., the effect of class belonging fuzzification of input features, the significance of fuzzy maximum membership value-based rule generation, interpretable knowledge extraction, and performance with the measures, such as Precision, Recall, KC, and KC_{std} performance of models using other classifiers, are also found to be true here for the DAELM-T-based DA algorithm.

4) Comparison of IRF-ELM Model With State-of-the-Art Methods: Comparative results of all nine (five fuzzy and four nonfuzzy) models, in terms of OA, KC, and OA_{std}, are illustrated in Table VII using the completely labeled samples scenario of RS data set. All these models have used the best possible likely optimum parameters for generating the results shown in Table VII. Fuzzy models performed better than the nonfuzzy model, as observed from Table VII, in an overall comparative scenario. Improvement of the fuzzy model is mainly attributed to its capability to address the information's impreciseness and uncertainty, typically seen in RS imageries. This characteristic of the fuzzy model also provides a better understanding of the domain that opens up broad scopes for improved accuracy in the final decision. In comparing IRF-ELM with IGMM-AL-DA, the latter approach is somehow deterministic grouping of training samples, unlike the IRF-ELM, which may not work with the random aspects in ELM architecture. Besides, IGMM-AL-DA dramatically depends on target- and source–domains' data structures, which is not a bothersome aspect in the ELM-based model used in the IRF-ELM. Moreover, IRF-ELM is computationally cheaper and holds improved generalization ability compared with IGMM-AL-DA because of its ELM architecture.

The performance of the IRF-ELM-DAPT (Model 5) is also compared with the ADDA model using LeNet-5 CNN architecture. Input to the CNN is a pixel vector of the RS data, and the output of the system is the label of the pixel vector. We have used CB fuzzified features of the training samples for the ADDA model as the input. In a comparison of performance (see Table VII), we found that the accuracy of ADDA was comparable with the IRF-ELM-DAPT model. Being a nonfuzzy model, ADDA could reach an equivalent level of accuracy with the CNN-based feature extraction and the adversarial DA approach. However, the computational complexity of ADDA is the primary concern in the present task. As expected, T_c value for the ADDA is 18.15 s, compared with around 0.25 using the proposed IRF-ELM-DAPT model. ADDA's time complexity is mainly attributed to three layers of convolution operations in CNN, a fully connected classifier with 500 iterations, and, finally, the adversarial architecture.

C. Supervised Land Cover Classification of RS Image

So far, we analyzed all five models' potentiality, including the IRF-ELM in the classification of RS data sets that worked with the framework DA. We make an experimental demonstration of the efficacy of models for the conventional supervised classification of LISS-II and SATIMAGE data sets. The whole data set is randomly divided into two groups for the experiment: training and others for testing the model. We have taken five different training–testing groups from the whole collection of samples; those are 20%, 40%, 60%, and 80% for training and the rest 80%, 60%, 40%, and 20% for testing, respectively.

1) IRS LISS-II Data Classification: The classification results of all five models (Model 1, Model 2, Model 3, Model 4, and Model 5) for the classification of LISS-II data set in terms of OA, OA_{std}, KC, KC_{std}, and T_c are obtained and depicted in Table VIII. The results are obtained using 200 rules because we verified experimentally that out of three different numbers of rules (200, 400, and 600), 200 rule-based networks outperformed others.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:05:59 UTC from IEEE Xplore. Restrictions apply.

TABLE VII

PERFORMANCE COMPARISON OF THE PROPOSED IRF-ELM-DAPT (IRF-ELM MODEL WITH DAPT-BASED DA ALGORITHM) MODEL WITH STATE-OF-THE-ART METHODS USING THE COMPLETELY LABELED SAMPLES SCENARIO OF RS DATA SET

Index	Non-fuzzy Model					Fuzzy Model				
	TCA-DA	GFK-DA	AE-DA	ELM-DAPT	ADDA	FR-DA	F-ELM-DAPT	TSK-DA	IGMM-AL-DA	IRF-ELM-DAPT
OA (%)	81.21	81.88	82.21	82.76	93.45	84.55	86.13	88.45	90.02	93.91
OA_{std} (%)	01.64	01.59	01.56	01.34	00.35	01.12	00.99	00.79	00.60	00.35
KC	0.7872	0.7982	0.8023	0.8221	0.9010	0.8431	0.8662	0.8761	0.8852	0.9030

TABLE VIII

COMPARISON OF MODELS USING LISS-II DATA IN THE SUPERVISED CLASSIFICATION FRAMEWORK FOR 200 NUMBER OF RULES

Model	Training sample (%)							
	20	40	60	80	80			
					OA_{std}	KC	KC_{std}	T_c (in sec.)
Model 1	65.13	72.34	75.56	82.13	1.83	0.701	0.713	1.631
Model 2	69.01	75.05	78.17	88.20	1.50	0.783	0.508	1.581
Model 3	71.41	79.83	82.01	89.13	1.25	0.914	0.446	1.562
Model 4	77.11	84.05	88.26	91.04	1.14	0.924	0.159	1.537
Model 5	80.70	87.34	92.11	94.03	0.67	0.951	0.036	1.051

TABLE IX

COMPARISON OF MODELS USING SATIMAGE DATA IN THE SUPERVISED CLASSIFICATION FRAMEWORK FOR 200 NUMBER OF RULES

Model	Training sample (%)							
	20	40	60	80	80			
					OA_{std}	KC	KC_{std}	T_c(in sec.)
Model 1	75.41	77.37	79.90	82.26	1.182	0.725	0.815	5.947
Model 2	77.04	79.34	81.45	84.78	1.635	0.771	0.622	5.271
Model 3	78.67	80.14	83.31	85.02	1.259	0.794	0.413	4.343
Model 4	80.11	82.63	84.45	86.14	1.036	0.824	0.245	5.013
Model 5	83.24	85.41	86.36	88.85	0.743	0.881	0.089	3.146

It is observed from the results shown in Table VIII that Model 1, using the conventional ELM with original features and fully connected network architecture, performed the worst among the rest. This is a similar trend of observation obtained by these models using the DA framework-based classification. All the indexes support the claim of superiority of the proposed IRF-ELM-based models in the classification of RS data for the conventional supervised framework.

One exciting and undeniable fact is coming out of the observation in Table VIII. The fact justifies the superiority of DA over a supervised framework. It is seen that with 60% of training samples of the LISS-II data set, supervised framework-based classification models are yielding inferior results with low OA and other indexes values, compared with the results obtained with DA frameworks-based models with 5% of training samples of LISS-II data as the target domain. The training set of DA-based models, indeed, contains 50% of the source–domain samples along with the target domain. However, our objective in this study is to develop DA-based models for classifying the target domain with very few labeled samples successfully. This observation signifies the objective and potentiality of a DA framework-based classification model.

2) SATIMAGE Data Set Classification: The IRF-ELM model's performance, along with the other four models, is also evaluated to classify this data set in the supervised classification framework. The classification accuracies in terms of OA, OA_{std}, KC, KC_{std}, and computation time (T_c) of all five models (Model 1–Model 5) are shown in Table IX. We observed that the 200 rule-based models yielded the best result compared with 400 and 600. There was no significant accuracy increment with the increase in the number of rules, but the computational cost increased.

It is seen from Table IX that Model 1 provided the OA value of 75.41 using 20% of training samples, which is the minimum. This value is 77.04 using Model 2 for 20% training samples. This improvement is expected with Model 2. Model 2 also takes less computation time compared with Model 1. This is because some of the weights of the ELM architecture of Model 2 are zero. This claim of Model 2 over Model 1 holds good for all training sets and other measurement indexes. The next level of improvement is obtained with Model 3 and Model 4, which have used data-dependent fuzzification of input features, i.e., CNB and CB methods. The observation reveals the class discriminative abilities of the CB and CNB fuzzification processes. Finally, among all models, Model 5 outperformed others. The claim is valid for all percentages of training sets of different data sets.

VII. CONCLUSION

The present study proposed an efficient classification model called IRF-ELM that worked with the DA-based framework. The model addressed the scarcity of labeled samples in RS imagery for land cover classification. We selected the conventional ELM as the base classifier for the proposed model because of its fast computing process and, at the same time, possesses universal approximation capability. The proposed model's prime objective was to classify an image in a particular domain with very few labeled samples by leveraging the already available known examples of a related domain image. The proposed DA model was designed using a supervised data distribution adaptation approach. With this approach, the proposed model brought data distribution of both source and target domains closer, which played a crucial role in the model's performance improvement. The contribution of this study to develop the model was fourfold. First, instead of crisp input, we used class belonging fuzzified inputs obtained by the data-dependent fuzzification method. Second, the interpretable ELM network was designed with the rules extracted from the source domain using the features' maximum membership value. Third, the random weights of ELM were replaced with deterministic binary weights of the rule matrix (R-matrix) obtained during the rule extraction. Fourth is the application of the proposed model for land cover classification of RS images

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:05:59 UTC from IEEE Xplore. Restrictions apply.

with very few labeled samples in a data set. We validated the model to classify the LISS-II data set using the LISS-III data set as the source–domain image.

We found the classification accuracy of the IRF-ELM as the highest among other F-ELM-based models. We evaluated the proposed model's performance in a supervised framework to classify SATIMAGE and LISS-II data sets. The proposed model justified its superiority in terms of various performance parameters in the classification of RS data sets in supervised and DA frameworks. We also realized the advantages of the ELM as the base classifier. This experimental study suggested that the proposed model is an efficient solution for RS image classification in the DA framework.

ACKNOWLEDGMENT

The authors would like to thank the Editor-in-Chief, Dr. Simon Yueh, and the Associate Editor and extend their gratitude to the anonymous reviewers for their constructive comments and suggestions to improve the overall quality and presentation of this article.

REFERENCES

[1] B. Tso and P. M. Mather, *Classification Methods for Remotely Sensed Data*, 2nd ed. Boca Raton, FL, USA: CRC Press, 2016.
[2] D. Tuia, C. Persello, and L. Bruzzone, "Domain adaptation for the classification of remote sensing data: An overview of recent advances," *IEEE Geosci. Remote Sens. Mag.*, vol. 4, no. 2, pp. 41–57, Jun. 2016.
[3] J. Yang, H. Zou, S. Cao, Z. Chen, and L. Xie, "MobileDA: Toward edge-domain adaptation," *IEEE Internet Things J.*, vol. 7, no. 8, pp. 6909–6918, Aug. 2020.
[4] G.-B. Huang, Q.-Y. Zhu, and C.-K. Siew, "Extreme learning machine: Theory and applications," *Neurocomputing*, vol. 70, nos. 1–3, pp. 489–501, Dec. 2006.
[5] S. Y. Wong, K. S. Yap, H. J. Yap, S. C. Tan, and S. W. Chang, "On equivalence of FIS and ELM for interpretable rule-based knowledge representation," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 26, no. 7, pp. 1417–1430, Jul. 2015.
[6] L. Zhang and D. Zhang, "Domain adaptation extreme learning machines for drift compensation in e-nose systems," *IEEE Trans. Instrum. Meas.*, vol. 64, no. 7, pp. 1790–1801, Jul. 2015.
[7] S. Xu, X. Mu, D. Chai, and S. Wang, "Adapting remote sensing to new domain with ELM parameter transfer," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 9, pp. 1618–1622, Sep. 2017.
[8] M. Volpi, G. Camps-Valls, and D. Tuia, "Spectral alignment of multi-temporal cross-sensor images with automated kernel canonical correlation analysis," *ISPRS J. Photogramm. Remote Sens.*, vol. 107, pp. 50–63, Sep. 2015.
[9] L. Duan, D. Xu, and I. Tsang, "Learning with augmented features for heterogeneous domain adaptation," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2012, pp. 1–8.
[10] H. Zuo, J. Lu, G. Zhang, and F. Liu, "Fuzzy transfer learning using an infinite Gaussian mixture model and active learning," *IEEE Trans. Fuzzy Syst.*, vol. 27, no. 2, pp. 291–303, Feb. 2019.
[11] H.-J. Rong, G.-B. Huang, N. Sundararajan, and P. Saratchandran, "Online sequential fuzzy extreme learning machine for function approximation and classification problems," *IEEE Trans. Syst., Man, Cybern. B. Cybern.*, vol. 39, no. 4, pp. 1067–1072, Aug. 2009.
[12] Z.-L. Sun, K.-F. Au, and T.-M. Choi, "A neuro-fuzzy inference system through integration of fuzzy logic and extreme learning machines," *IEEE Trans. Syst., Man, Cybern. B. Cybern.*, vol. 37, no. 5, pp. 1321–1331, Oct. 2007.
[13] N. K. Kasabov, "Learning fuzzy rules and approximate reasoning in fuzzy neural networks and hybrid systems," *Fuzzy Sets Syst.*, vol. 82, no. 2, pp. 135–149, Sep. 1996.
[14] S. K. Meher and D. A. Kumar, "Ensemble of adaptive rule-based granular neural network classifiers for multispectral remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 5, pp. 2222–2231, May 2015.
[15] L. A. Zadeh, "Fuzzy sets," *Inf. Control*, vol. 8, no. 3, pp. 338–353, Jun. 1965.
[16] S. K. Pal and S. Mitra, "Multilayer perceptron, fuzzy sets, and classification," *IEEE Trans. Neural Netw.*, vol. 3, no. 5, pp. 683–697, 1992.
[17] S. K. Meher, "Explicit rough-fuzzy pattern classification model," *Pattern Recognit. Lett.*, vol. 36, pp. 54–61, 2014.
[18] S. K. Pal, S. K. Meher, and S. Dutta, "Class-dependent rough-fuzzy granular space, dispersion index and classification," *Pattern Recognit.*, vol. 45, no. 7, pp. 2690–2707, Jul. 2012.
[19] C. M. Bishop, *Pattern Recognition and Machine Learning*. New York, NY, USA: Springer-Verlag, 2006.
[20] A. Samat, P. Du, J. Y. Li, L. Cheng, and S. Liu, "E2LMs: Ensemble extreme learning machines for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 4, pp. 1060–1069, Apr. 2014.
[21] S. M. Salaken, A. Khosravi, T. Nguyen, and S. Nahavandi, "Extreme learning machine based transfer learning algorithms: A survey," *Neurocomputing*, vol. 267, pp. 516–524, Dec. 2017.
[22] V. Behbood, J. Lu, and G. Zhang, "Fuzzy refinement domain adaptation for long term prediction in banking ecosystem," *IEEE Trans. Ind. Informat.*, vol. 10, no. 2, pp. 1637–1646, May 2014.
[23] C. Yang, Z. Deng, K.-S. Choi, and S. Wang, "Takagi–Sugeno–Kang transfer learning fuzzy logic system for the adaptive recognition of epileptic electroencephalogram signals," *IEEE Trans. Fuzzy Syst.*, vol. 24, no. 5, pp. 1079–1094, Oct. 2016.
[24] J. Deng, Z. Zhang, F. Eyben, and B. Schuller, "Autoencoder-based unsupervised domain adaptation for speech emotion recognition," *IEEE Signal Process. Lett.*, vol. 21, no. 9, pp. 1068–1072, Sep. 2014.
[25] B. Gong, Y. Shi, F. Sha, and K. Grauman, "Geodesic flow kernel for unsupervised domain adaptation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2012, pp. 16–21.
[26] S. J. Pan, I. W. Tsang, J. T. Kwok, and Q. Yang, "Domain adaptation via transfer component analysis," *IEEE Trans. Neural Netw.*, vol. 22, no. 2, pp. 199–210, Feb. 2011.
[27] E. Tzeng, J. Hoffman, K. Saenko, and T. Darrell, "Adversarial discriminative domain adaptation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2962–2971.
[28] Y. Lecun, L. Bottou, Y. Bengio, and P. Haffner, "Gradient-based learning applied to document recognition," *Proc. IEEE*, vol. 86, no. 11, pp. 2278–2324, Oct. 1998.
[29] J. Cohen, "A coefficient of agreement for nominal scale," *Educ. Psycol. Meas.*, vol. 20, pp. 37–46, 1960.
[30] S. K. Pal, A. Ghosh, and B. U. Shankar, "Segmentation of remotely sensed images with fuzzy thresholding, and quantitative evaluation," *Int. J. Remote Sens.*, vol. 21, no. 11, pp. 2269–2300, Jan. 2000.
[31] D. L. Davies and D. W. Bouldin, "A cluster separation measure," *IEEE Trans. Pattern Anal. Mach. Intell.*, vols. PAMI–1, no. 2, pp. 224–227, Apr. 1979.
[32] ISRO. (Aug. 2011). *Bhuvan Data Set*. [Online]. Available: http://bhuvan.rsc.gov.in



Saroj K. Meher (Senior Member, IEEE) received the Ph.D. degree in electronics engineering from the National Institute of Technology, Rourkela, India, in 2003.

He is an Associate Professor with SSIU, Indian Statistical Institute, Bangalore Centre, India. His research interests include pattern recognition, granular computing, and computational intelligence.



Neeta S. Kothari (Graduate Student Member, IEEE) received the B.E. degree (Hons.) from Rajiv Gandhi Proudyogiki Vishwavidyalaya (RGPV), Bhopal, India, in 2009, and the M.E. degree from Devi Ahilya Vishwavidyalaya (DAVV), Indore, India, in 2013, both in electronics engineering.

She is a Visiting Scholar with Systems Science and Informatics Unit (SSIU), Indian Statistical Institute, Bangalore Centre, India. Her research interests include pattern recognition, granular neural networks, and remote sensing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:05:59 UTC from IEEE Xplore. Restrictions apply.